**Math515 – Financial Mathematics II**

**Spring 2014**

Homework 4

1. Prove the the following properties of step functions:

   - Linearity:

   $$\int_a^b (\alpha\phi(t,\omega) + \beta\psi(t,\omega))\, dW_t(\omega) = \alpha \int_a^b \phi(t,\omega)\, dW_t(\omega) + \beta \int_a^b \psi(t,\omega)\, dW_t(\omega);$$

   - Union:

   $$\int_a^b \phi(t,\omega)\, dW_t(\omega) + \int_b^c \phi(t,\omega)\, dW_t(\omega) = \int_a^c \phi(t,\omega)\, dW_t(\omega).$$

2. Compute the Ito integral $\int_0^T (W_t^3 - t)\, dW_t$, then verify the martingale property.

3. Consider the Ito integral

   $$I(\omega) = \int_0^T W_t(\omega)\, dW_t(\omega) = \frac{1}{2}[(W_T(\omega))^2 - T].$$

   Define $I_n(\omega) = \sum_{j=0}^{n-1} W_j \Delta W_j$ with a uniform partition $t_j = j\Delta t$ ($\Delta t = T/n$) of $[0,T]$.

   (a) Show that

   $$I - I_n = \frac{1}{2}\sum_{j=0}^{n-1}[(\Delta W_j)^2 - \Delta t].$$

   (b) Then show that $dist(I, I_n) = E[(I - I_n)^2]$ tends to zero as $n$ goes to infinity.

4. (a) Consider $d(W_t(\omega))^k$ by Ito's formula and the martingale property of Ito integrals to deduce

   $$E[(W_T(\omega))^k] = \frac{1}{2}k(k-1)\int_0^T E[(W_t(\omega))^{k-2}]dt,$$

   for $k \geq 2$.

   (b) Based on the result from (a) to determine $E[(W_T(\omega))^2]$, $E[(W_T(\omega))^4]$ and $E[(W_T(\omega))^6]$.